Tutorial 4

1. **Integration of Functions**

   (a) Write a function that implements the simple trapezoid rule, and use it to calculate the following integrals:

$$\int_1^5 x^2 \mathrm{d}x, \tag{1}$$

$$\int_0^\pi \sin(x)\mathrm{d}x. \tag{2}$$

   (b) Now implement a function that uses Simpson's method to calculate the integrals. First, call the trapezoid method twice - once using N points, and once with $2N$ points, then combine them to get Simpson's method as in the lecture.

   (c) Now extend this routine into a Romberg integration routine by following the algorithm showed in the lecture. Compare the results given by the three methods, and see which one gives the most accurate estimation. Use $N = 8$ and a 6th order Romberg method (that is, 6 initial approximations).

   (d) Calculate the value of

$$\int_0^{10} \left[3\exp(-2x) + \left(\frac{x}{10}\right)^4\right] \mathrm{d}x \tag{3}$$

   for N equally spaced points in the given range for x. Use the trapezoid and Simpson's rule for $N = 10$, 100, 1000 and 10000, as well as the Romberg method with $N = 10$ and $m = 2$, 4 and 6. Calculate the exact value of the integral, or a highly precise estimation using e.g. WolframAlpha, and compare the results. Which method gave the best estimation and for which value of N? What is the dependence of N with the error of calculation? (error = Exact - Estimated value)

Solution:

(a) The analytic solutions to those integrals are $41.\bar{3}$ and 2, respectively. Note that you can implement the trapezoid rule without any `for` loop. Using the array of equally spaced x_i, you can compute in one command the corresponding $f(x_i)$. Then you `sum` the $f(x_i)$ corresponding to all internal x_i (i.e., excluding the two extreme values) weighting them by h and add the two extreme values weighted by $h/2$.

(b) As we know analytically a-priori the sequence of multiplicative factors for each $f(x_i)$, one could implement also the Simpson's rule without recurring to `for` loops. One possible way would be to initially identify those indices whose $f(x_i)$ need to be multiplied by 4 and those which instead need to be multiplied by 2 (`mod` can be helpful here combined with fancy/advanced indexing), paying attention to the extremes, and combining everything.

(c) For Romberg's method you can e.g. first loop downwards in the 'pyramid' from the slides, calculating the integral using progressively smaller values of h. Then, you need another loop to combine the integral estimates using the recurrence relation.

(d) The analytic solution of the integral is $7/2 - 3/2\exp(-20) \approx 3.4999999969082696...$ Figure 1 shows how the integration error behaves as a function of the number of points or order for the three methods.

(e) The analytic solution is $-3/64(6\sqrt{14} - 25\ln 5 - 382\ln(3+\sqrt{14})) \approx 35.0043196050514..$ (from WolframAlpha). Figure 2 shows how the integration error behaves as a function of the number of points or order for the three methods.

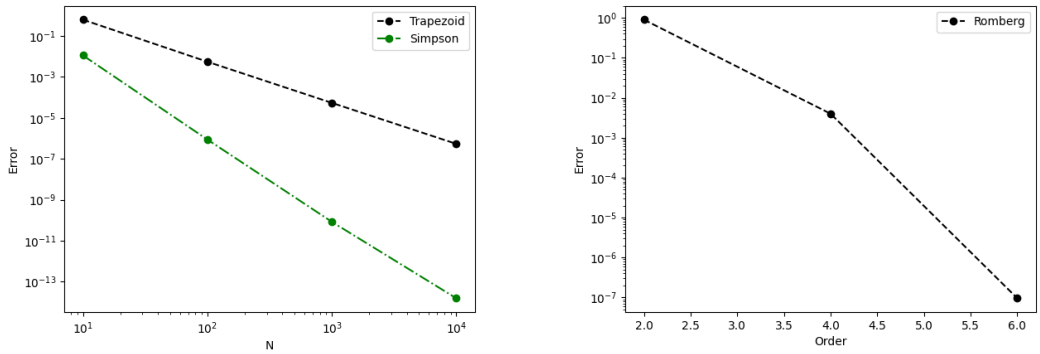

Figure 1: **Left panel:** Integration errors for the trapezoid and Simpson's methods as a function of the number of points; **Right panel:** Integration error for the Romberg method as a function of order.

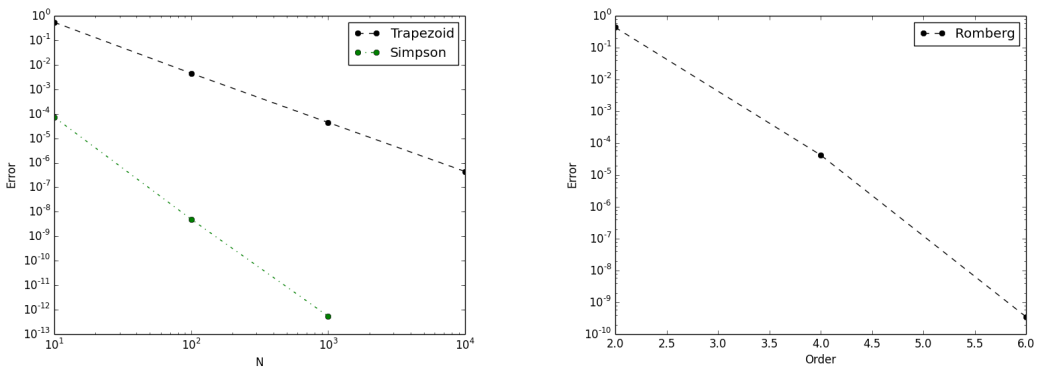

Figure 2: **Left panel:** Integration errors for the trapezoid and Simpson's methods as a function of the number of points; **Right panel:** Integration error for the Romberg method as a function of order.

2. **Improper Integrals**

(a) Consider the following integral:

$$\int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x}}. \tag{4}$$

What kind of integral is this, and can you use your functions from the previous exercise to solve it, or do you need to change anything in your algorithms?

(b) The Gamma function has various uses in astronomy to define or estimate properties of objects. One way to define the complete Gamma function is by using the following convergent improper integral:

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} \mathrm{d}x. \tag{5}$$

Estimate $\Gamma(6)$. (Hint: Divide the integral in parts with convenient limits first.)

(c) The complete Gamma function is equal to $\Gamma(z) = (z-1)!$ for integer z. Calculate $\Gamma(6)$ using this formula and compare with your estimation.

(d) The Incomplete Gamma Function is another variety of Gamma function and has multiple uses

in astronomy too. One way of defining it is as follows:

$$\Gamma(z, x) = \int_x^\infty t^{z-1} e^{-t} \mathrm{d}t. \tag{6}$$

Estimate $\Gamma(6, 2.5)$.

Solution:

(a) This is an integral that is undefined at one of the limits, so it is not possible to calculate its value using a closed interval, and using a open method is needed. Any integration method is possible, as long as it's open. Romberg integration results in 2.

(b) This integral can be solved in different ways. One option is to use a change of variables. Another option is to use an open interval or using an adaptive dx for the integration. The result should be around 120.

(c) Use implicit programming, see e.g. the following website:
`https://www.programiz.com/python-programming/examples/factorial`

(d) Again, you can use the same approaches as (b), the result should be around 114.43 when using numerical methods.